" EXPLORING CANCER TRENDS IN INDIA: A STATISTICAL APPROACH TO GENDER AND ZONAL VARIATION "

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Abstract: Ever wondered how numbers quietly narrate the stories that shape a nation's health? Cancer statistics do exactly that—revealing patterns that often go unnoticed yet influence public health in profound ways. Behind every dataset lies a tale of resilience, disparity, and urgent need. This study steps into that silent world of data, decoding the evolving landscape of cancer in India through a statistical lens.

As per the NCRP 2020 dataset, the analysis examined demographic patterns across zones by integrating gender distribution, age structure, and regional variation. The chi-square test confirmed that there is a difference between the cancer cases of males and females, demonstrating a meaningful statistical association at the 5% significance level.

Covariance and correlation analysis showed an extremely strong positive linear relationship between males and females population counts of 0.9986.

This pattern is further supported by regression analysis, indicating a highly predictable and stable demographic relationship between genders.

The people of all age group are not at the same risk of developing of developing cancer. The cancer cases are not equally distributed across all zones. The west zone has recorded the highest incidence of cancer cases in India.

By exploring zonal variation and gender disparities, this study reveals critical gaps and patterns that can guide future prevention strategies and healthcare planning.

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Introduction:

India is undergoing rapid demographic, environmental, and lifestyle transitions, all of which are significantly reshaping the nation's health profile. Cancer is a significant global health concern, with India ranking second in Asia and third in the world in terms of cancer incidence. Regular monitoring and updates on cancer statistics are vital for assessing the impact and burden of the disease and the effectiveness of cancer control measures (https://pmc.ncbi.nlm.nih.gov/articles/PMC12368690/). With rising life expectancy, changing dietary habits, increased pollution levels, and shifts in lifestyle patterns, cancer incidence has a grown steady increase across the country. The expanding population and improvement in diagnostic facilities have further contributed to higher cancer detection rates in recent years. Cancer is a group of diseases involving abnormal cell growth with the potential to invade or spread to other parts of the body. (https://en.wikipedia.org/wiki/Cancer)

The distribution of cancer in India is highly uneven across regions, age groups and genders. Northern and northeastern states often report higher incidence due to lifestyle practices, tobacco use and environmental exposures, while urban cities experience increased cases linked to sedentary habits, processed food consumption, and pollution. In contrast, rural regions continue to struggle with late diagnosis due to limited access to screening, healthcare facilities, and awareness. this variation reflects India's Diverse socio-economic and cultural landscape, influencing both risk exposure and healthcare accessibility.

Despite significant advancements in medical infrastructure and national programs, challenges such as delayed diagnosis, low health-literacy, and socio-economic disparities persist. Strengthening screening programs, promoting awareness, and improving access to affordable treatment are essential to reduce growth cancer burden. understanding cancer trends across India is therefore crucial for guiding prevention strategies, enhancing early detection, and improving overall public health planning.

From Numbers to Narratives: Statistics often appear cold and distant, but in this context, they carry emotion and urgency. Each graph, rate, and comparison sheds light on unequal burdens—why certain zones report higher cases, why some cancers dominate among women while others affect men disproportionately, and how socio-economic and regional frameworks influence detection and outcomes. These insights push beyond simple description, offering direction for policy, awareness, and resource allocation.

From Patterns to Prevention: Cancer trends are more than counts on a chart—they are reflections of lifestyle shifts, environmental exposures, and genetic predispositions. By uncovering gender-based differences and zonal disparities, this study highlights the hidden dynamics influencing cancer incidence across the country. It combines rigorous data analysis with public health insight to reveal how

demographic and geographical factors converge to shape cancer risk.

Cancer is not just a disease—it is a complex societal challenge influenced by culture, access, and awareness. Understanding its trajectory requires more than medical knowledge; it demands a statistical journey through India's diverse landscape.

Objectives

The major objective of the research is to:

- To examine the spatial distribution of cancer incidence across various zones in India.
- To assess whether geographical zone of residence influences the probability of individuals being diagnosed cancer positive.
- To analyse whether gender plays a role in the number of cancers in India.
- To analyse the age group most at risk of developing cancer.

Methods and Materials

The research is entirely based on secondary data, sourced from the National Centre for Disease Informatics and Research (NCDIR), a government authorized body responsible for maintaining standardized and validated disease records. Using NCDIR data ensured reliability, authenticity and consistency as the information is systematically collected and widely used for epidemiological research and public health planning. The use od secondary data also allowed access to large-scale population level information across regions, age groups and genders which would not be feasible through primary data collection.

The dataset is organised into six major zones: North Zone, South Zone, North East Zone, West Zone, East Zone and Central Zone. Each zone comprises of multiple states, including Manipur, Mizoram, Sikkim, Meghalaya and Tripura in the North East. Punjab and Delhi in the North. Telangana, Tamil Nadu and Karnataka in the South. West Bengal in the East. Gujarat and Maharashtra in the West. Madhya Pradesh in the Central Zoen. Within each state, cancer cases are categorized by gender and geographic variations.

Statistical Techniques:

1. Correlation Analysis:
   Pearsons Correlation Coefficient Formula:

   $$r = \frac{n\sum XY - \sum X \sum Y}{(n\sum X^2 - (\sum X)^2)(n\sum Y^2 - (\sum Y)^2)}$$

   where,

   n= Number of data points

   X and Y are the two variables being compared

   This is the most convenient form for computing r from raw data (Gun, Gupta and Dasgupta, 2019)

2. Simple Linear Regression:
   The major purpose of regression is to explore the dependence of one variable on another. In SLR, the mean of a random variable, Y id modelled as a function of another observed variable X (Casella et.ai.,2007)
   Regression Equation:

   $$Y = \beta_0 + \beta_1 X + \varepsilon$$

where,

Y is the dependent variable

β_0 is the intercept

β_1 is the coefficient of the independent variables X

ε is the error term

3. Hypothesis Testing and $\mathbf{X^2}$ statistics:

A test of a statistical hypothesis is a two- action decision problem after the experimental sample values have been obtained, two-actions being the acceptance or rejection of the hypothesis under consideration (Gupta & Kapoor, 2021)

Null Hypothesis:

The null hypothesis states there is no significant relationship or effect between the variables studied.

Alternate Hypothesis:

The alternative hypothesis is the assumption that there is a meaningful and measurable effect or relationship between the variables being investigated.

The chi-square distribution may be defined as the 'Sum of the squares of independent normally distributed variables with zero mean and unit variances.' (Gupta & Kapoor, 2021). It is given by the formula:

$$\chi^2 = \sum \frac{(Observed\ Frequency - Expected\ Frequency)^2}{Expected\ Frequency}$$

Data Analysis

1. **Which zone has the highest incidence of cancer cases in India?**

Zones	Total
Central Zone	41254260
East Zone	89541595
North East Zone	54540323
North Zone	157220198
South Zone	103419407
West Zone	223443845

Table 1.1: Zonal distribution of Cancer Cases

Graphical Depiction:

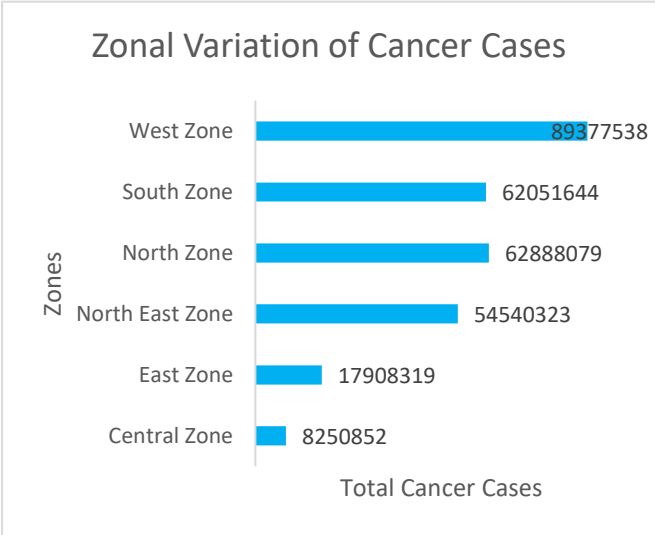


Fig 1.1: Graphical Representation of Zonal distribution of Cancer Cases

Interpretation:
Fig 1.1 illustrates the zonal distribution of cancer cases revealing geographical disparities. The West Zone bears the highest cancer burden with over 89 million, followed closely by the North and South Zones at around 62.8 million and 62 million respectively, indicating a concentration of cases in these regions. The Central Zone reports the lowest cancer burden among the zones with over 8 million cancer cases.

2. **Which state in each zone has the highest cancer cases?**

i. **North East Zone**

Region	Total
MANIPUR	15778153
MIZORAM	5862530
SIKKIM	3324050
MEGHALAYA	10019395
TRIPURA	19556195

Table 1.2: Cancer cases in Regions of North East Zone

Graphical Depiction:

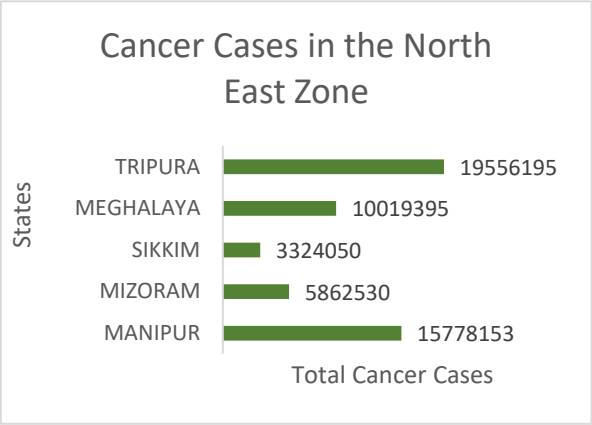


Fig 1.2: Graphical Representation of Cancer Cases in Regions of North East Zone

Interpretation:

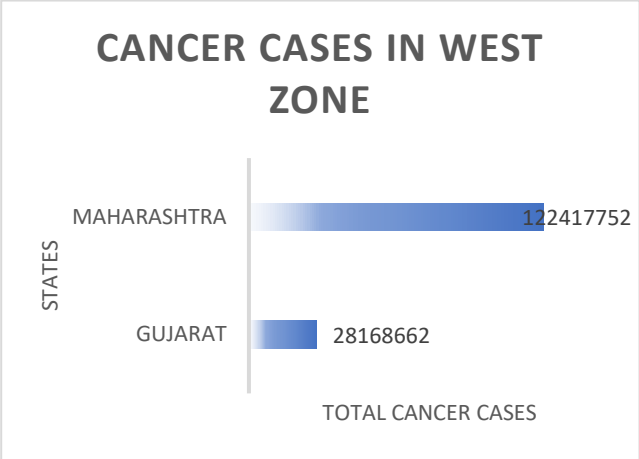
Fig 1.2 shows incidence of cancer cases in the north east zone consisting of 5 states namely: Tripura, Meghalaya, Sikkim, Mizoram and Manipur. The distribution of cancer cases across the North Eastern state's shows notable variation. Tripura records the highest number of registered cases that is about 19 million, reflecting the greatest cancer which also shows a relatively high incidence. Meghalaya occupies a moderate position with a mid-range number of cases of over 10 million. In contrast, Mizoram and Sikkim reported comparatively lower cancer cases at around 5million and 3 million respectively.

West Zone:

Region	Total
GUJARAT	28168662
MAHARASHTRA	122417752

Table 1.3: Cancer cases in Regions of West Zone

Graphical Depiction:



Interpretation:

The west zone consists of 2 states namely: Maharashtra and Gujarat. The distribution of cancer cases across centres shows clear variation. Maharashtra records the over 122 million cancer cases, indicating the greatest cancer burden.

Central Zone:

Region	Total
MADHYA PRADESH	8250852

Graphical Depiction:

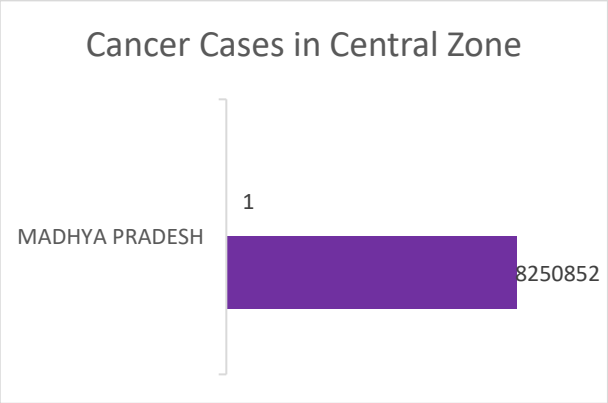


Fig 1.4: Graphical Representation of Cancer Cases in Regions of Central Zone

Interpretation and Conclusion:

The central zone has only 1 state named Madhya Pradesh. Madhya Pradesh reports a substantial cancer burden with around 8.2 million registered cases, indicating significant pressure on the states healthcare system

East Zone:

Region	Total
West Bengal	17908319

Table 1.4: Cancer cases in Regions of Central Zone

Graphical Depiction:

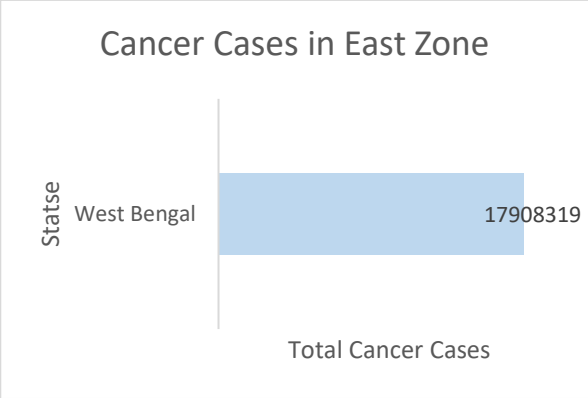


Fig 1.5: Graphical Representation of Cancer Cases in Regions of East Zone

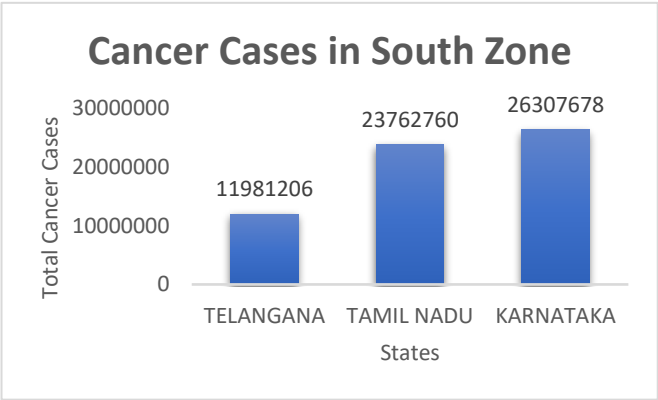
Interpretation:
The data for East Zone consists of 1 state namely West Bengal. It recorded over 17 million cancer cases, making it a region with significant patient volume.

ii. South Zone:

Region	Total
TELANGANA	11981206
TAMIL NADU	23762760
KARNATAKA	26307678

Table 1.5: Cancer cases in Regions of South Zone

Graphical Depiction:



Interpretation:

The south zone consists of 3 states namely Tamil Nadu, Telangana and Karnataka. The distribution of cancer cases across the three metropolitan centres shows clear variation. Karnataka reports the highest number of cases at around 26 million, In contrast, Telangana records significantly fewer cases at over 11 million.

iii. North Zone:

Region	Total
PUNJAB	10065059
DELHI	51923020

Table 1.7: Cancer cases in Regions of North Zone

Graphical Depiction:

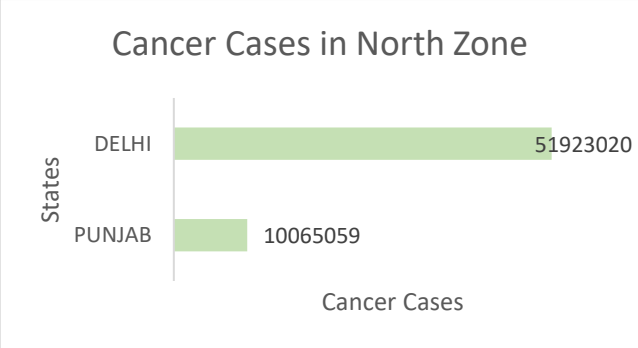


Fig 1.7: Graphical Representation of Cancer Cases in Regions of North Zone

Interpretation:
The north zone comprises of 2 states namely Delhi and Punjab. The comparison between Delhi and Punjab reveals a pronounced disparity in cancer burden. Delhi records a significantly higher number of cases, with over 51 million registered, positioning it a major cancer care hub with exceptionally high patient volume. In contrast, Punjab reports around 10 million cases.

3. Which age group id highly prone to cancer?

Age Group	Total
00-04	23456241
05-09	24373027
10-14	25635098.5
15-19	26876491
20-24	30442587

25-29	29320648.5
30-34	25468697
35-39	23097636
40-44	19860908
45-49	17065502.5
50-54	13595898
55-59	10586263.5
60-64	8912146
65-69	5981238
70-74	4558818.5
75+	5785554.5

Table 1.8: Age wise distribution of Cancer Cases

Graphical Depiction:

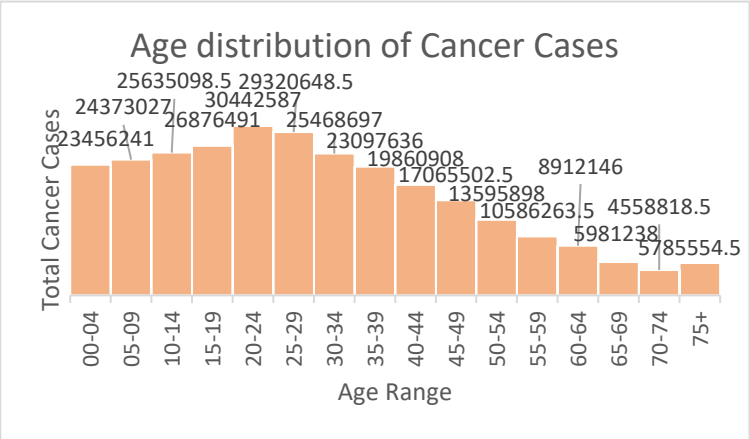


Fig 1.8: Graphical Representation of Age wise distribution of Cancer Cases

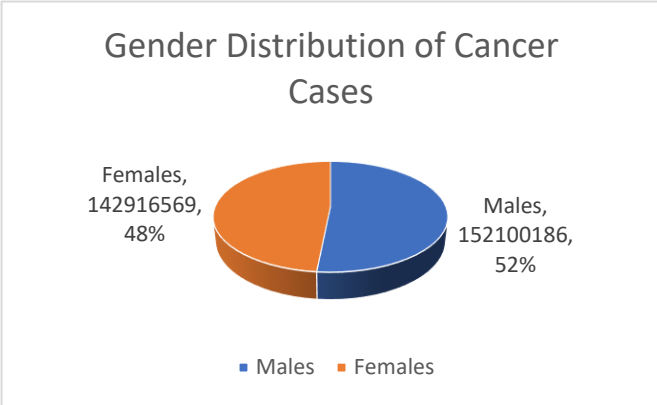
Interpretation:
The age wise distribution of cancer cases shows a clear and distinct pattern. Cancer incidence increases steadily from early childhood to young adulthood, peaking in the 20-24 age group. High case numbers persist through the 25-29 and 30-34 age groups, after which a gradual decline is observed. From the 40-44 age group onward, the decrease becomes more pronounced, with the lowest incidence recorded among the oldest age groups (70-74 and 75+).

4. **Which gender is most at risk of developing cancer?**

Gender	Total
Males	152100186
Females	142916569

Table 1.9: Total cancer cases against the gender

Graphical Depiction:



Interpretation:
The nearly equal distribution of cancer cases between males and females indicates that cancer represents a shared health burden rather than being concentrated in a single gender. Males correspond to 52% of the total cancer cases while female cancer cases contribute to 48% of the cases.

5. Region wise gender analysis of cancer patients

Zones	Males	Females
Central Zone	4280917	3969935
East Zone	9270946	8637373
North East Zone	27541445	26998878
North Zone	32929570	29958509
South Zone	31641265	30410379
West Zone	46436043	42941495

Table 2.0: Gender wise distribution of Cancer cases in every zone

Graphical Depiction:

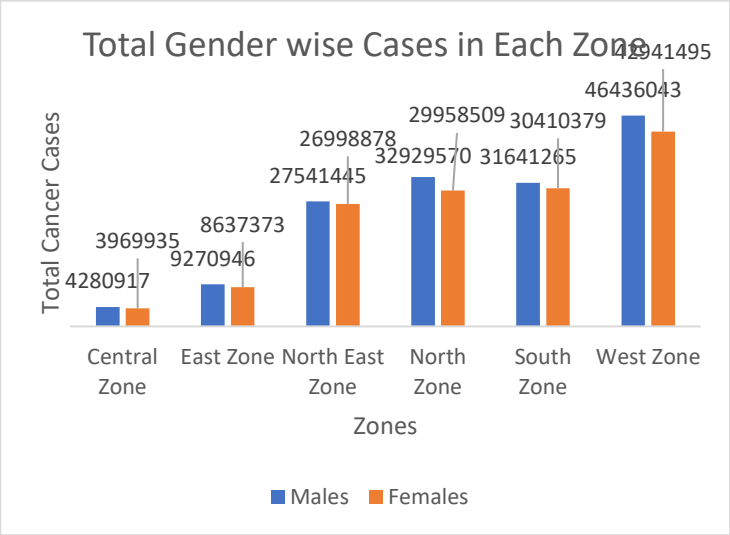


Fig 2.0: Graphical Representation of Gender wise distribution of Cancer cases in every zone

Interpretation:

Figure 2.0 highlights clear zonal variations in cancer cases across genders. Overall, male cases are marginally higher than female cases in all zones, indicating a broadly similar cancer burden between genders with regional differences in magnitude. The Central and East Zones report relatively lower-case counts, with the East Zone uniquely showing slightly higher female cases. Cancer incidence rises sharply in the North-East, North and South Zones where both genders record high and closely aligned case number of cases for males and females likely influenced by higher population density and better diagnostic coverage.

1. **To analyse whether gender plays a role in the number of cancers cases in India.**

Zones	Males (X)	Females (Y)
Central Zone	4280917	3969935
East Zone	9270946	8637373
North East Zone	27541445	26998878
North Zone	32929570	29958509
South Zone	31641265	30410379
West Zone	46436043	42941495

Table 2.1: Representation of number of cancer cases among males and females in every zone

Hypothesis:

- H_0 (Null Hypothesis)
  There is no difference between the cancer cases of males and females
- H_1 (Alternative Hypothesis):
  There is a difference between the cancer cases of males and females

Observed Frequency:

Zones	Males (X)	Females (Y)	Total
Central	4280917	3969935	8250852
East	9270946	8637373	17908319
North East	27541445	26998878	54540323
North	32929570	29958509	62888079
South	31641265	30410379	62051644
West	46436043	42941495	89377538
Total	152100186	142916569	295016755

Table 2.2: Observed Frequency of number of cancer cases among males and females in every zone

Expected Frequency:

Zones	Males (X)	Females (Y)	Total
Central	4253846.951	3997005.05	8250852
East	9232894.758	8675424.24	17908319
North East	28119058.09	26421264.9	54540323
North	32422865.32	30465213.7	62888079
South	31991629.07	30060014.9	62051644
West	46079891.82	43297646.2	89377538
Total	152100186	142916569	295016755

Table 2.3: Expected Frequency of number of cancer cases among males and females in every zone

Final Calculation:

The chi-square test statistics at **α=0.05** and **degree of freedom= 5** is given by:

$$X^2 = \sum \frac{(Observed\ Frequency - Expected\ Frequency)^2}{Expected\ Frequency}$$

X^2 calculated= 55121.49618

Using the χ^2 table, we get:

χ^2 tabulated= 11.07

Conclusion:

Since χ^2tabulated is less than X^2 calculated that is 11.07 < 55121.49618. Thus, we reject the null hypothesis (H0).

Regression Analysis:

$$\bar{x} = \frac{\sum x}{n} = 25350031$$

$$\bar{y} = \frac{\sum y}{n} = 23819428.17$$

N = 6

$$\text{Cov(X, Y)} = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{n-1} = 232099405607035$$

$$\text{Var (X)} = \frac{\Sigma\,(x-\bar{x})^2}{n-1} = 249779154652208$$

$$\beta_1 = \frac{\text{Cov(X,Y)}}{\text{Var (X)}} = 0.93$$

$$\beta_0 = \bar{y} - \beta_1\,\bar{x} = 263710.9756$$

The Regression Equation

The regression equation is given by:

Y = 263710.9756 + 0.93X

This means that every 1 increase in the male cancer cases there the cancer case of female increases by 0.93. This means the rise in cancer cases among female population is proportional to male population across all zones.

Correlation Analysis:

$$r = \frac{\sum(x-\bar{x})*(y-\bar{y})}{\sqrt{(\sum(x-\bar{x})^2)*(\sum(y-\bar{y})^2)}} = 0.998596566$$

Conclusion:

The variable x and y are highly correlation. This means a high degree of correlation can be seen in the cancer cases of males and females and that is 0.9986.

2. **To assess whether geographical zone of residence influences the probability of individuals being diagnosed cancer positive.**

Zones	Total
Central Zone	41254260
East Zone	89541595
North East Zone	54540323
North Zone	157220197.5
South Zone	103419406.7
West Zone	223443845

Table 2.4: Zonal representation of total cancer cases

Hypothesis

- H_0 (Null Hypothesis)
  Cancer cases are equally distributed across all zones
- H_1 (Alternative Hypothesis):
  Cancer cases are not equally distributed across all zones

Final Calculation:

Zones	Total	Expected	Chi Term
Central	41254260	111569937.9	44315652.1
East	89541595	111569937.9	4349270.95
North East	54540323	111569937.9	29151015.4
North	157220197.5	111569937.9	18678384.6
South	103419406.7	111569937.9	595421.67
West	223443845	111569937.9	112178705

Table 2.5: χ^2 calculation

The χ^2 term in table 2.4 is calculated using:

$$\chi^2 = \frac{(\textbf{Observed Frequency} - \textbf{Expected Frequency})\ \hat{}\ \mathbf{2}}{\textbf{Expected Frequency}}$$

X^2 calculated = 209268449.2

Using the chi square table, we get

χ^2tabulated= 11.07

Conclusion:

Since χ^2tabulated is less than X^2 calculated that is 11.07 < 209268449.2. Thus, we reject the null hypothesis (H0). Hence, Cancer cases are not equally distributed across all zones

3. **To analyse the age group most at risk of developing cancer.**

Age Group	Total
00-04	23456241
05-09	24373027
10-14	25635098.5
15-19	26876491
20-24	30442587
25-29	29320648.5
30-34	25468697
35-39	23097636
40-44	19860908
45-49	17065502.5
50-54	13595898
55-59	10586263.5
60-64	8912146
65-69	5981238
70-74	4558818.5
75+	5785554.5

Table 2.6: Age wise distribution of cancer cases

Hypothesis:

- H_0 (Null Hypothesis)
  People of all age group are at the same risk of developing cancer
- H_1 (Alternative Hypothesis):

  People of all age group are not at the same risk of developing cancer

Age Group	Total	Expected	Chi Term
00-04	23456241	18438547.2	1365468
05-09	24373027	18438547.2	1910023
10-14	25635098.5	18438547.2	2808809
15-19	26876491	18438547.2	3861416
20-24	30442587	18438547.2	7814985
25-29	29320648.5	18438547.2	6422422
30-34	25468697	18438547.2	2680418
35-39	23097636	18438547.2	1177268
40-44	19860908	18438547.2	109721.8
45-49	17065502.5	18438547.2	102245.1
50-54	13595898	18438547.2	1271860
55-59	10586263.5	18438547.2	3343992
60-64	8912146	18438547.2	4921880
65-69	5981238	18438547.2	8416311
70-74	4558818.5	18438547.2	10448050
75+	5785554.5	18438547.2	8682800

Table 2.7: χ^2 calculation

Calculations:

The χ^2 term in table 2.7 is calculated using:

$$\chi^2 = \frac{(\text{Observed Frequency} - \text{Expected Frequency})\ \hat{}\ 2}{\textit{Expected Frequency}}$$

X^2 calculated = 65337669.4

Using the chi square table, we get

χ^2tabulated= 11.07

Since χ^2tabulated is less than X^2 calculated that is 11.07 < 65337669.4. Thus, we reject the null hypothesis (H0). Hence, People of all age group are not at the same risk of developing cancer.

Results and Discussions:

The study narrates that there are many zones with higher with higher number of cancer patients in comparison to other. Similarly, the cancer cases also differ with gender groups and age-groups. The findings emphasize the need for region-specific public health strategies, improved cancer surveillance systems, and equitable access to screening and treatment facilities. These insights can assist policymakers and healthcare planners in designing targeted interventions to effectively manage and reduce the cancer burden across the country.

Overall, the findings highlight uneven distribution of cancer cases, with certain zones experiencing disproportionately higher prevalence underscoring the need for zone specific health planning and resource allocation. There is a significant interstate disparity emphasizing the need for state specific cancer prevention, screening and treatment strategies tailored to the actual burden of diseases. These disparities highlight the need for centre specific strategies with enhanced diagnostic capacity and treatment support in high burden centres, alongside strengthened outreach efforts in lower reporting regions. Some high burden regions, the figure remains considerable and may be influenced by the state's large population, delayed diagnosis, limited awareness, and uneven access to

specialized cancer care, particularly between urban and rural areas. the findings indicate a disproportionately higher cancer burden among younger and middle-aged adults, highlighting the importance of targeted awareness programs, early screening and preventive innervations focused on younger age groups.

Overall, the findings emphasize the need to strengthen healthcare infrastructure, expand early public awareness initiative to effectively manage the cancer burden and improve health outcomes across the state.

Conclusion:

This study provides a detailed look at cancer incidence patterns across different regions, states, age groups, and genders in India. It uses secondary data from the National Centre for Disease Informatics and Research. The findings show a highly uneven distribution of cancer cases. Regions like the West, North, and South report significantly higher rates compared to the Central and East zones. These geographic differences point to factors like population density, healthcare infrastructure, diagnostic reach, and reporting methods affecting the cancer trends.

The analysis by gender shows a nearly equal distribution of cancer cases between males and females in most areas. This suggests that cancer is a common public health issue rather than one that affects just one gender. This balance highlights the need for gender-neutral approaches in cancer prevention, screening, and treatment, while still paying attention to specific risks for each gender. Age-related patterns show that cancer rates are particularly high among younger and middle-aged adults, especially in early adulthood, followed by a gradual decline in older groups. This underscores the importance of early screening, awareness campaigns, and lifestyle changes aimed at younger populations.

Comparisons between states and zones reveal significant differences in cancer burden. Metropolitan areas and healthcare hubs report higher numbers, likely due to better diagnostic access and patients coming from nearby regions. Statistical analyses, such as chi-square tests and correlation studies, confirm strong links between demographic factors and cancer distribution, reinforcing that the patterns in the data are structured and not random.

However, the study has limitations due to its dependence on secondary data, incomplete district-level coverage, and potential reporting inconsistencies across states and regions. Still, the findings provide important evidence for policymakers, healthcare planners, and public health officials. Improving cancer surveillance systems, ensuring fair access to screening and treatment, and creating targeted interventions for specific areas will be crucial in lowering the cancer burden and enhancing health outcomes for the population. Overall, this study helps clarify cancer distribution patterns in India and serves as a foundation for future research and policy development. By transforming raw data into meaningful narratives, the study contributes to a deeper understanding of cancer's impact on India—helping illuminate the path toward early detection, equitable healthcare, and informed national strategy.

References

- Gupta, S.C. and Kapoor, V.K. (2021), 13th edition, Fundamentals of Mathematical Statistics, Sultan Chand & Sons, p.15.3
- Casella, George and Berger, Roger L. (2007). 2nd edition, Statistical Inference, Cengage India Private Limited
- Gun, A.M, Gupta, M.K., and Dasgupta, B. (2019), 3rd

Edition, Fundamentals of Statistics. World Press Private Limited. P. 303, 11.4 – correlation coefficient

- Gupta, S.C. and Kapoor, V.K. (2021), 14th edition, Fundamentals of Applied Statistics, Sultan Chand & Sons, Hypothesis Testing, p.18.2
- https://ncdirindia.org/All_Reports/
- https://en.wikipedia.org/wiki/Cancer
- https://pmc.ncbi.nlm.nih.gov/articles/PMC12368690/